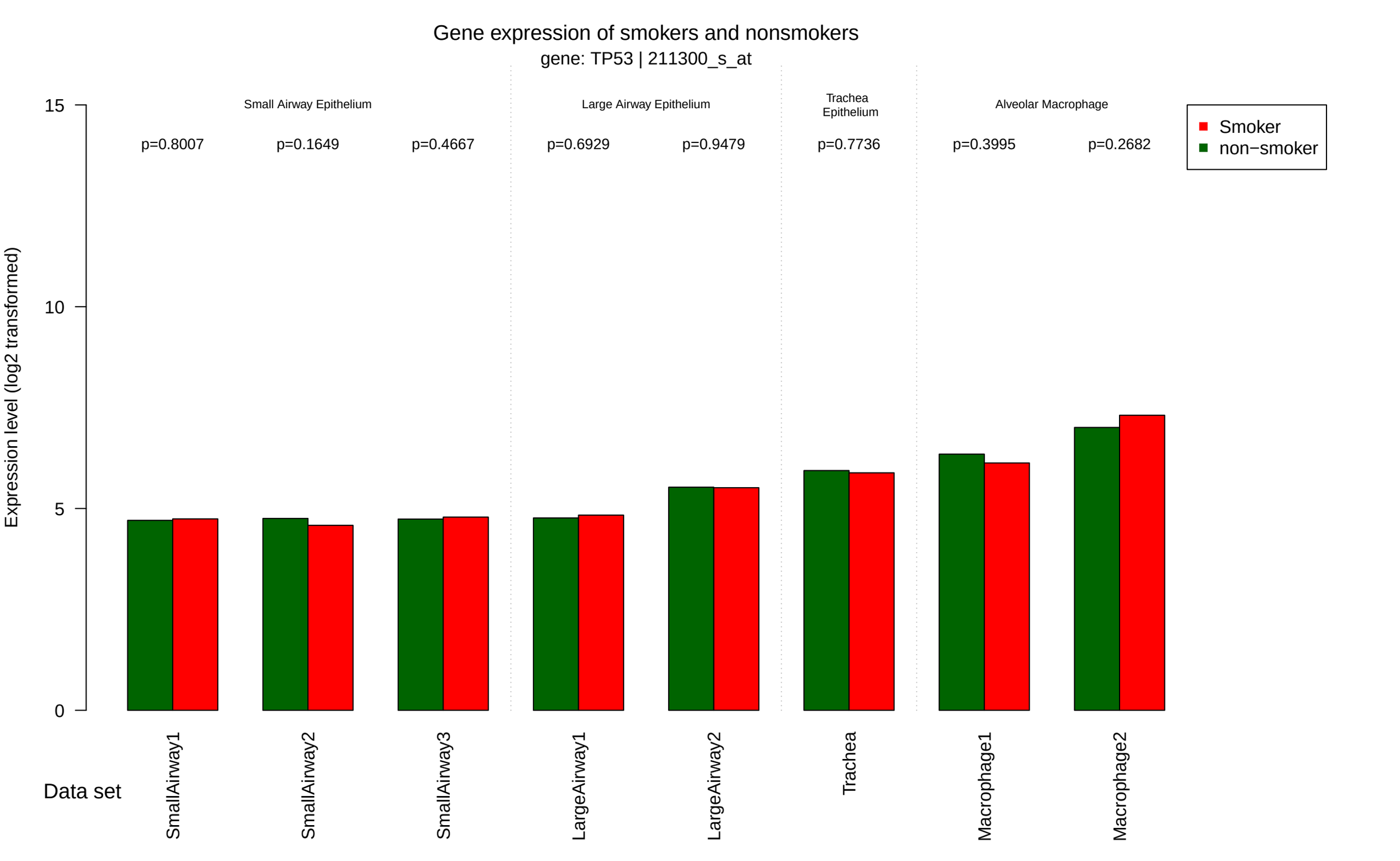
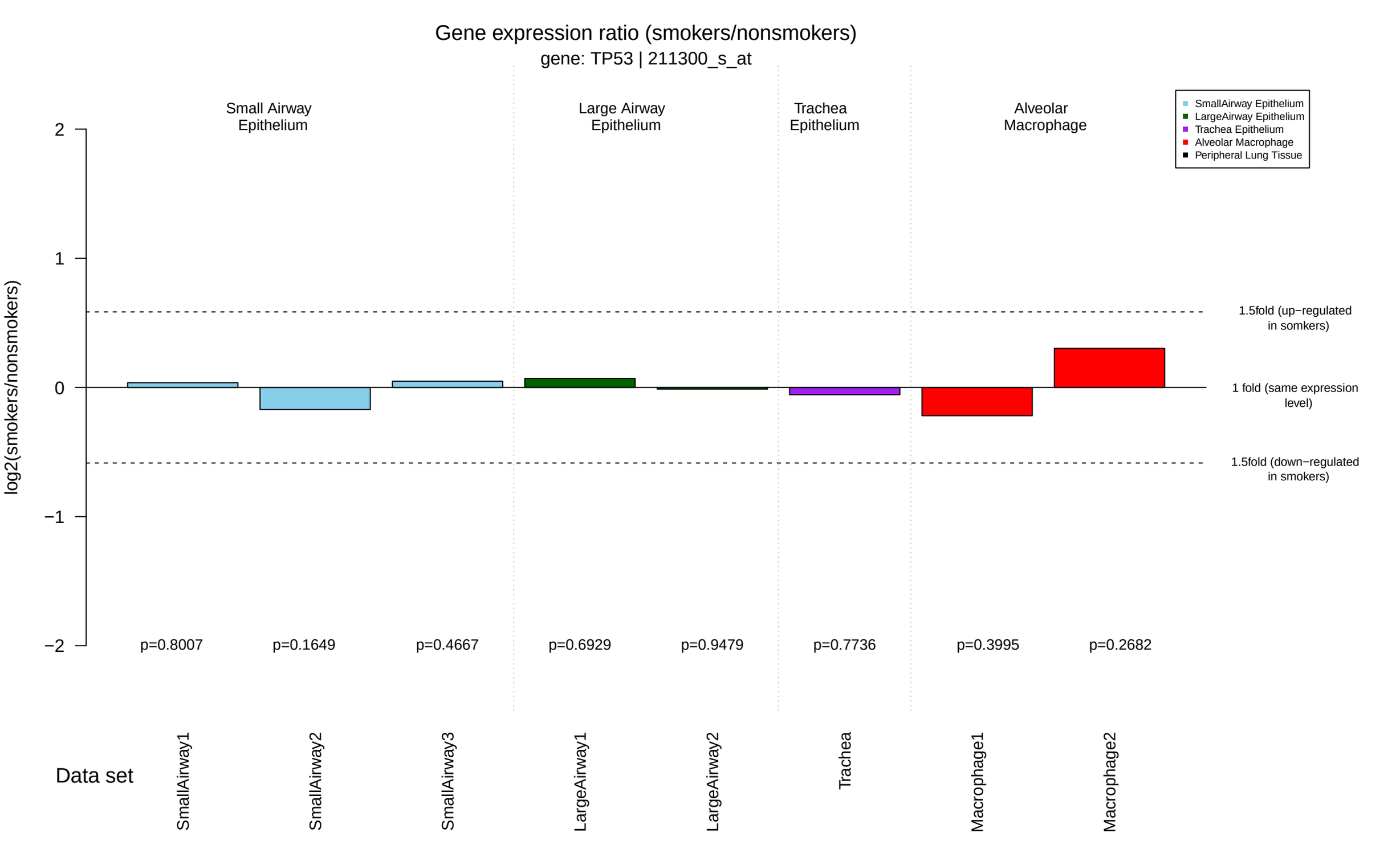
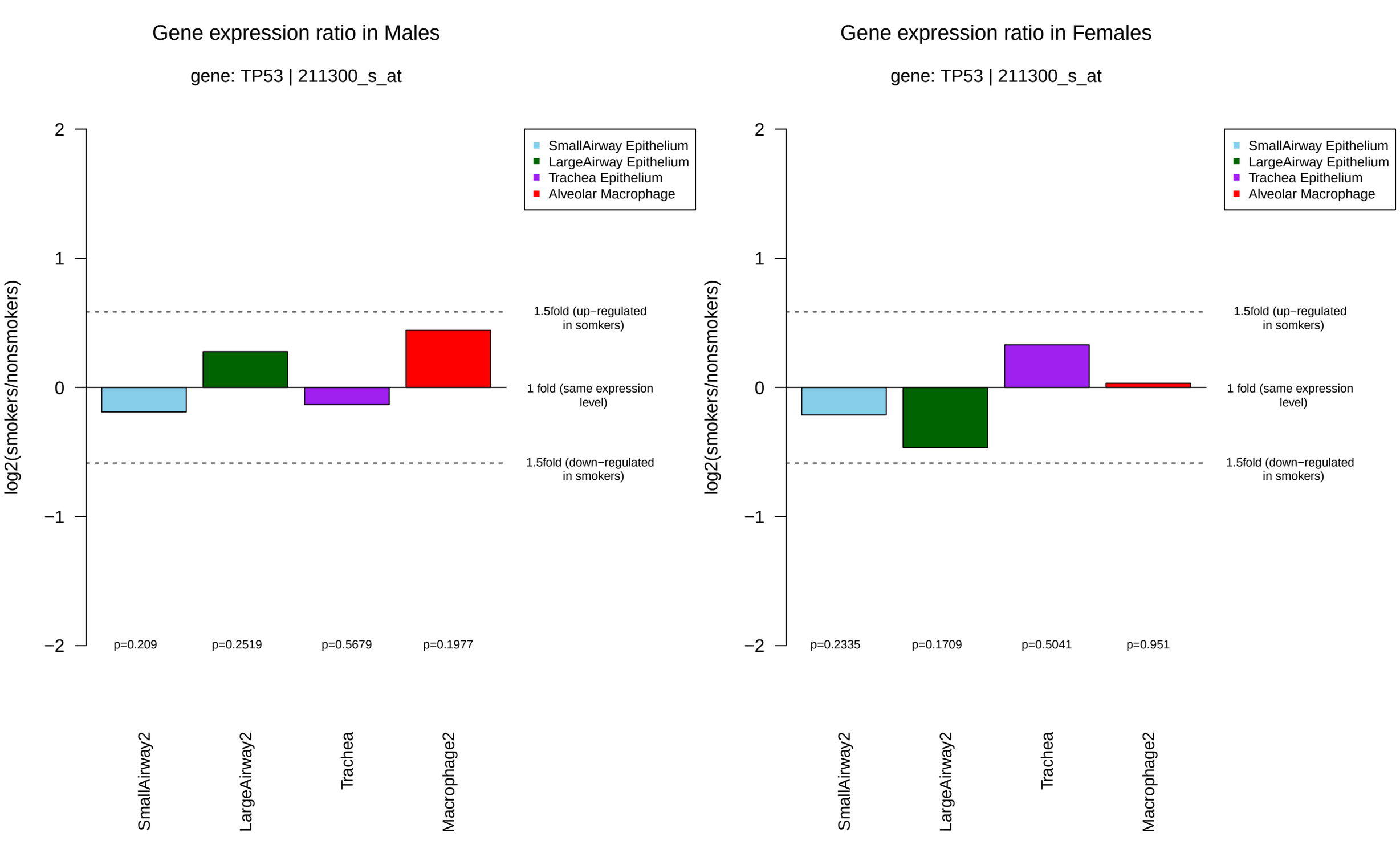
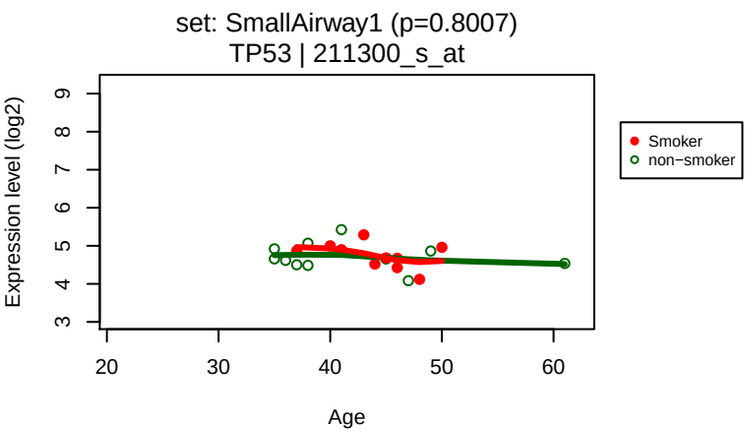
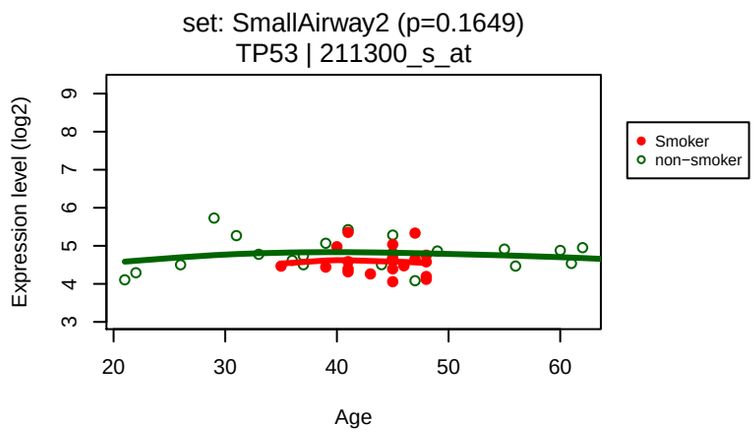
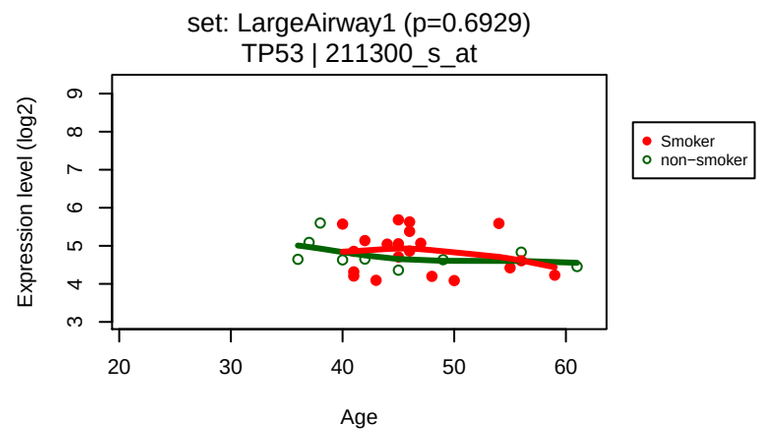

Gene expression of smokers and nonsmokers
gene: TP53 | 211300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8007
p=0.1649
p=0.4667
p=0.6929
p=0.9479
p=0.7736
p=0.3995
p=0.2682
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

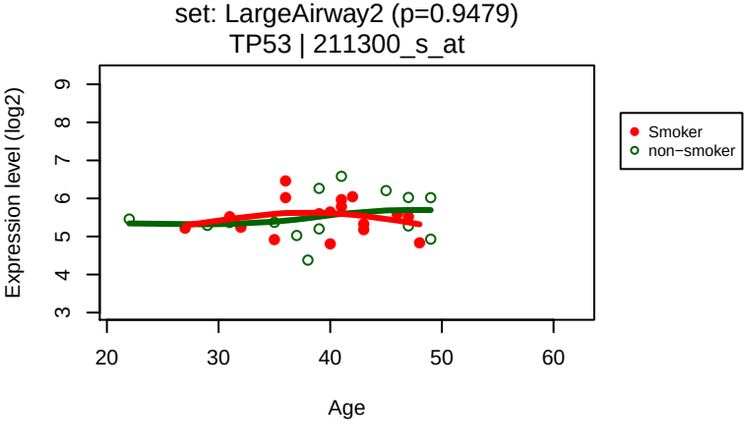

Gene expression ratio (smokers/nonsmokers)
gene: TP53 | 211300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8007
p=0.1649
p=0.4667
p=0.6929
p=0.9479
p=0.7736
p=0.3995
p=0.2682
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

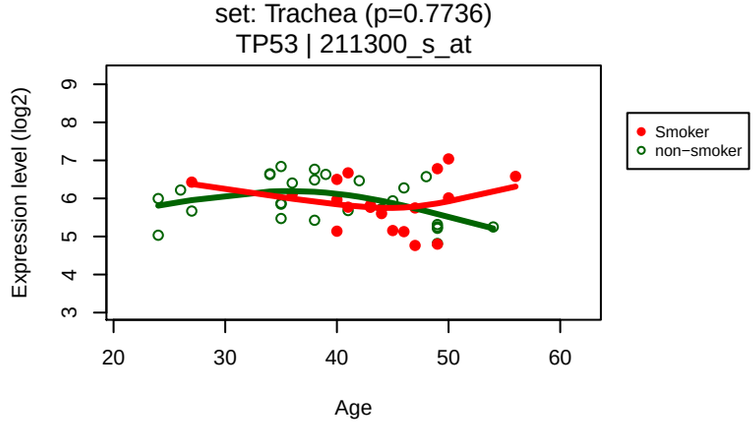

Gene expression ratio in Males
gene: TP53 | 211300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.209
p=0.2519
p=0.5679
p=0.1977
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TP53 | 211300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2335
p=0.1709
p=0.5041
p=0.951
SmallAirway2
LargeAirway2
Trachea
Macrophage2

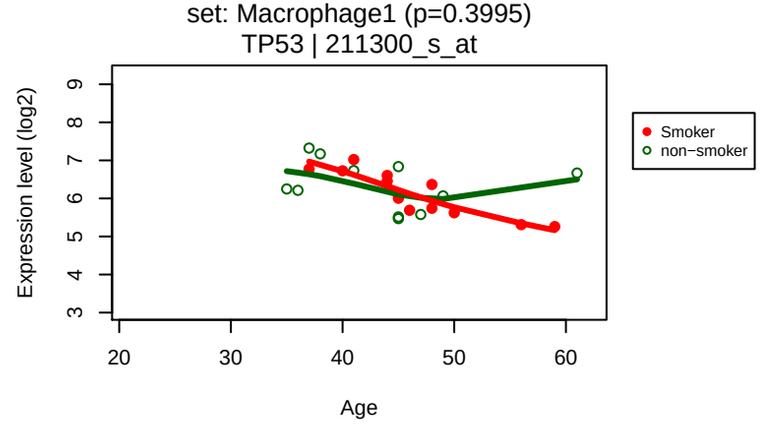

set: SmallAirway1 (p=0.8007)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker

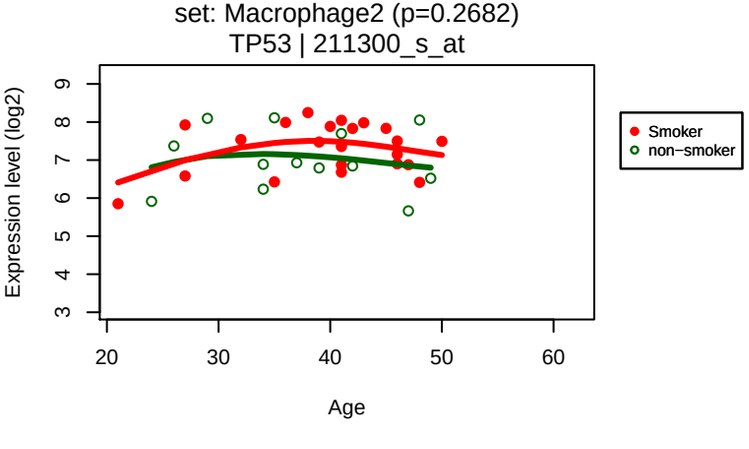

set: SmallAirway2 (p=0.1649)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker

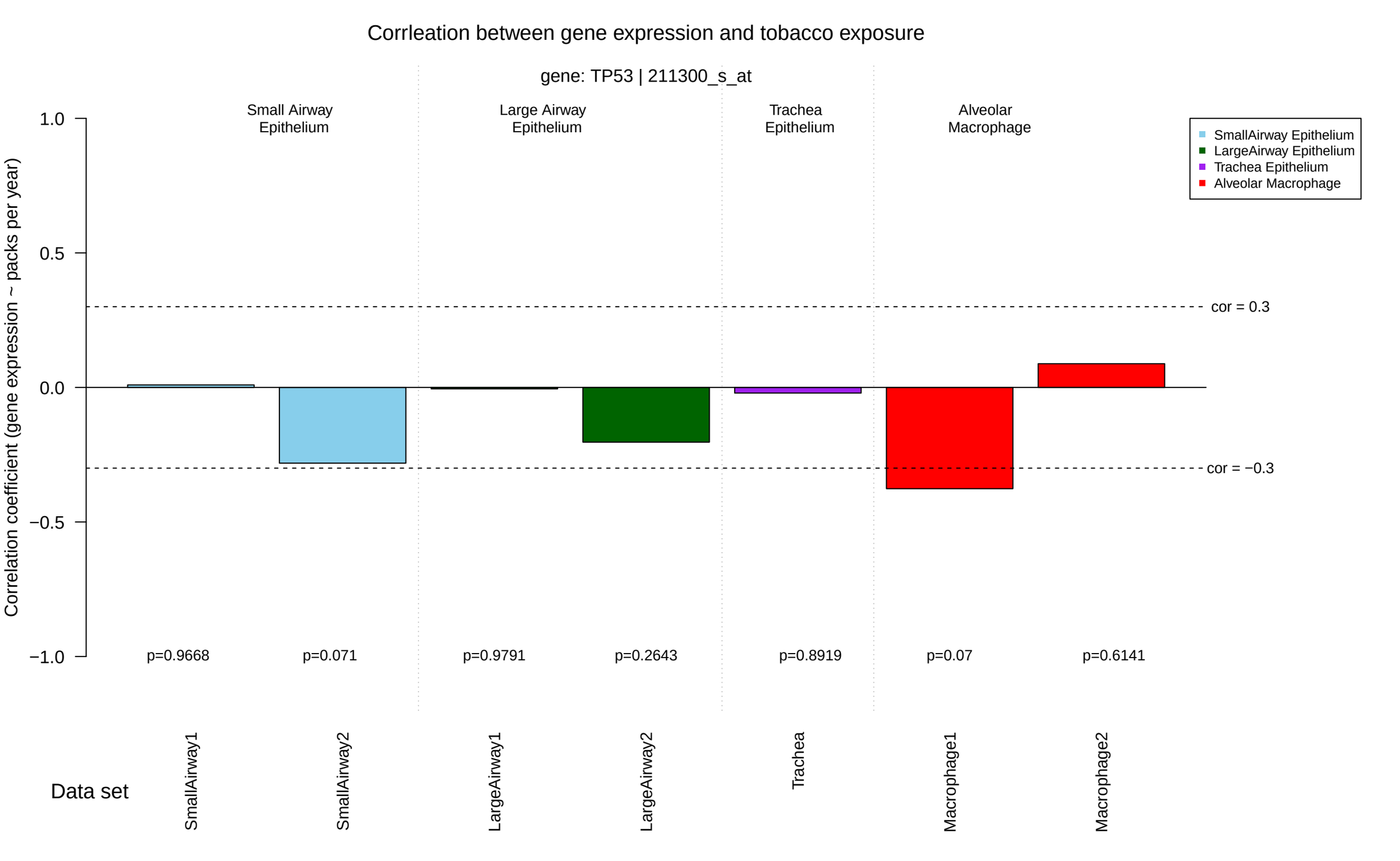

set: LargeAirway1 (p=0.6929)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9479)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7736)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3995)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2682)
TP53 | 211300_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TP53 | 211300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9668
p=0.071
p=0.9791
p=0.2643
p=0.8919
p=0.07
p=0.6141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2